

Problem 1: Missing Values

Problem Found:

Several columns in the combined dataset contain missing values (NaN), including checkout information, member information, and book information.

1. Affected Columns

The missing values were found in the following columns:

- checkout_id
- return_date
- first_name
- last_name
- grade
- neighborhood
- membership_status
- join_date
- publication_year

2. Affected Rows

The missing values affected different numbers of rows:

- Checkout_id : 26 rows
- Return_date : 91 rows
- First_name : 5 rows
- Last_name : 5 rows
- Grade : 41 rows
- Neighborhood : 5 rows
- Membership_status : 5 rows
- Join_date : 11 rows
- Publication_year : 35 rows initially (after dropping duplicates 33 rows)

3. How I Handled It

I investigated the missing values by checking the available source data. When the missing information could not be reliably recovered from the sources, I **kept the values as missing (NaN)** rather than filling them with estimated values.

4. Why I Handled It This Way

I did not fill the missing values with assumptions or statistical values such as mean, median, or mode because these values represent identifiers, dates, or specific information about members/books. Using estimated values could introduce inaccurate information and affect the reliability of the analysis.

Problem 2: Duplicate Checkouts

Problem Found:

Some checkout records were duplicated in the combined dataset, meaning the same checkout transaction appeared more than once.

1. Affected Columns

The duplicate records were identified by comparing the complete checkout records across the dataset, including the checkout information and the related member and book details.

2. Affected Rows

A total of **(8 duplicate rows)** were identified.

3. How I Handled It

I examined the duplicate records to determine whether they were **true duplicates or only similar records**. The 8 records were confirmed as true duplicates, so they were removed using `drop_duplicates()`.

4. Why I Handled It This Way

The duplicate records represented repeated copies of the same checkout information. Keeping them would cause the same checkout transaction to be counted more than once and could lead to inaccurate analysis.

Problem 3: Data Inconsistency

Problem Found:

Some values were written in different formats even though they represented the same information. For example, Active and active, or Zamalek and zamalek.

1. Affected Columns

The inconsistencies were found mainly in:

- neighborhood
- membership_status

Examples included:

- Zamalek / zamalek
- Nasr City / NASR CITY
- Heliopolis / HELIOPOLIS
- Maadi / Maadi (extra space)
- Active / active
- Inactive / inactive

2. Affected Rows

The affected rows were the records containing these differently formatted values. The exact number was identified during the inconsistency check.

3. How I Handled It

I standardized the values by:

- Using consistent capitalization.
- Removing unnecessary spaces.
- Converting active → Active.
- Converting inactive → Inactive.
- Standardizing neighborhood names to one consistent format.

4. Why I Handled It This Way

I standardized these values so that the same category would not be treated as different categories during filtering, grouping, and analysis. This improves the consistency and accuracy of the dataset without deleting valid records.

Problem 4: Checkouts With No Matching Member

Problem Found:

Some checkout records contained a `member_id` that did not have a corresponding record in the `members` table. These records were identified as orphan checkout records

1. Affected Column

The issue was identified in the (`member_id`) column, where some checkout records contained member IDs that had no matching record in the (`members` table.)

2. Affected Rows

A total of **5 checkout rows** were affected, corresponding to

3 unique member IDs: 1104, 1201, and 1150.

3. How I Handled It

I compared the `member_id` values in the checkout records with the IDs available in the `members` table. The affected checkout records were **kept**, while the unavailable member details remained missing.

4. Why I Handled It This Way

The checkout records contain valid information about book transactions, so deleting them would result in losing useful data. Since no matching member records were available in the source, I could not reliably recover the missing member details without creating inaccurate information.